



A Machine Learning-ready Data Set for Exoplanet Atmospheric Retrieval

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Abstract

Atmospheric retrieval is a modeling technique used to determine a planet’s atmosphere’s temperature and composition from spectral data. The retrieved atmospheric composition can provide understanding into the surface fluxes of gaseous species necessary to maintain the stability of that atmosphere, leading to insights into the geological as well as biological processes active on the planet. Among exoplanets, rocky terrestrial ones are of particular interest because of their theoretical habitability. Atmospheric retrieval is both time consuming and computationally intensive. Traditional retrieval methods involve the use of complex algorithms that generate numerous atmospheric models. These models are then compared to observational data, and a posterior distribution is constructed to determine the most likely value and associated uncertainty for each model parameter. Runtimes scale with the number of model parameters, and when many molecular species are considered, become prohibitively long. The issue will become especially prohibitive as the number of detected exoplanets will grow tremendously in the near future. Machine learning (ML) offers a way to reduce the time to perform a retrieval by orders of magnitude, given a sufficient data set to train with. Here we present a large data set of 3,112,620 synthetic planetary systems generated with our Intelligent Exoplanet Atmospheric Retrieval framework based on the NASA Planetary Spectrum Generator. The data set contains the parameters defining each planetary system and the simulated spectra of stellar, planetary and noise components. The data set was designed to enable the first ML retrieval model for rocky terrestrial exoplanets, and it is publicly available through the NASA Exoplanet Archive.

Unified Astronomy Thesaurus concepts: [Exoplanet atmospheres \(487\)](#); [Biosignatures \(2018\)](#); [Astrobiology \(74\)](#); [Habitable zone \(696\)](#); [Computational methods \(1965\)](#); [Distributed computing \(1971\)](#); [Astronomy databases \(83\)](#)

1. Introduction

In recent years, the number of detected exoplanets has grown to over 5000 (J. L. Christiansen 2018). To date, characterizing the atmospheres of these exoplanets has been limited by the sensitivity and spectral resolution of available instruments. Terrestrial exoplanets, the most difficult class of exoplanets to characterize, offer the opportunity to detect combinations of molecules indicative of life (biosignatures; E. W. Schwieterman et al. 2018). The detection of biosignatures is driving future telescope designs such as the Habitable Worlds Observatory concept (National Academies of Sciences, Engineering, and Medicine 2023). While the James Webb Space Telescope (JWST) will be able to detect key molecules that together can constitute a biosignature, this represents the limit of JWST’s capability over its lifetime and depends on the lack of high-altitude aerosols (L. Kaltenegger 2017; J. Krissansen-Totton et al. 2018; J. Lustig-Yaeger et al. 2019; D. Gasman et al. 2022).

The detection of exoplanets can be direct when the signal emitted by the planet’s surface and atmosphere are measured, or indirect, which relies on the effect that the planet’s presence exerts on its host star. Among indirect observations, the transit method consists of detecting stellar signal variations due to the transit of the planet. The direct measurement of the apparent planetary radius as a function of wavelength contains information about the atmospheric structure and composition of the planet, as the molecular species present in the atmosphere at the day–night terminator absorb differing amounts of stellar flux at particular wavelengths (I. J. M. Crossfield 2015).

In coronagraphic observations, a coronagraph suppresses the stellar emission, directly measuring a combination of the planet’s emission and stellar flux reflected by the planet’s atmosphere and/or surface (Y. Fujii et al. 2018). With sufficient measurements at multiple wavelengths, the atmospheric conditions can be retrieved with some degree of uncertainty via Bayesian inverse modeling (see the review by N. Madhusudhan 2018).

The accuracy of the retrieved constraints on atmospheric properties is limited by the spectral coverage and resolution of the measurements. Presently, such techniques can be reasonably applied to only a small subset of the total number of observed exoplanets, as most have only been observed by one



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or two instruments using wide-band photometry. This problem is exacerbated as the retrieval model grows in complexity, and as the growth in dimensionality renders traditional retrieval approaches infeasible.

Machine learning (ML) approaches to exoplanet atmospheric retrieval (e.g., P. Márquez-Neila et al. 2018; F. Soboczenski et al. 2018; T. Zingales & I. P. Waldmann 2018; A. D. Cobb et al. 2019; T. K. Johnsen et al. 2020; M. D. Himes et al. 2022) have been demonstrated to significantly reduce the computational costs of retrieval compared to traditional non-ML methods. This computational cost reduction allows dozens to hundreds of retrievals to be carried out in the time it takes a traditional algorithm to perform a single retrieval. This enables retrieval studies to determine the possible retrieval constraints over a parameter space of interest (e.g., observing time, instrument/telescope parameters, planetary/stellar properties).

While catalogs of confirmed exoplanets and their published properties exist (e.g., D. Bashi et al. 2018; J. L. Christiansen 2018), the lack of detailed atmospheric models and high-resolution spectra prohibits their use for the problem of retrieval as well as other related problems such as biosignature detection or atmospheric composition studies. ML depends on the size and the quality of the data set used for training, validation, and testing (R. L. Figueroa et al. 2012). Currently, only a few terrestrial exoplanets are known, and this data set would not be large enough for ML even if each had been observed with high-resolution spectroscopy. Developing retrieval methods based on state-of-the-art ML techniques often require large-scale training data of modeled spectra and their corresponding atmospheric properties, but few such data sets exist publicly at present. For example, the ATMO grids of hot Jupiter transmission spectra (J. M. Goyal et al. 2018, 2019, 2020) and the Sonora grids of brown dwarf spectra (T. Karalidi et al. 2021; M. S. Marley et al. 2021; C. V. Morley et al. 2024; S. Mukherjee et al. 2024) are publicly available for investigations related to those classes of objects, the latter of which have been used for an ML retrieval study (A. Lueber et al. 2023). The public availability of such data sets is crucial for the community both to easily compare different methodologies as well as explore new ML approaches to retrieval-related problems.

Here we present a data set of 3,112,620 synthetic terrestrial exoplanet spectra generated using the Intelligent exoplanet Atmosphere Retrieval (INARA) framework. The data set focuses on planets in the optimistically habitable zone as defined by R. K. Kopparapu et al. (2013). The complete data set is publicly available to the community at the NASA Exoplanet Archive through a browsable web interface or as bulk downloads. In Section 3, we detail INARA’s data generation methods. Section 4 describes the online web interface to interact with the INARA data set, and data availability. In Section 5, we showcase a practical application of the data set. Specifically, we demonstrate how a neural network can be trained to accurately forecast the properties of an exoplanet’s atmosphere based on its observed spectrum. Finally, in Section 6 we present conclusions, data and code availability.

2. Objectives

In this work, we concentrate on the development and release of a comprehensive data set tailored for atmospheric retrieval of artificial rocky terrestrial exoplanets. Exoplanets of this category represent pivotal targets for current and upcoming

space missions. Through this data set, our primary objective is to enable ML-based approaches that can drastically reduce the computational cost of, and enable novel approaches to, atmospheric retrieval. The core components of our work include: (i) generating a large and homogeneous data set of rocky exoplanets that can be used for ML-based approaches; (ii) implementing efficient ML algorithms to demonstrate the use of our data set; and (iii) provisioning high-performance computing resources to facilitate synthetic data generation and the training and evaluation of ML models. While the current data set focuses on rocky terrestrial exoplanets, it is intended for future application to exoplanets potentially observable with advanced instruments. More generally, the INARA framework can be extended to produce data sets for other planet types (e.g., hot Jupiters) currently observable, and can be used for exploratory studies to determine constraints under assumed telescope properties and noise models (see Section 3 for details, including the simulated telescope setup and reference to ongoing efforts such as the Habitable Worlds Observatory).

To construct our data set, we employ the NASA Planetary Spectrum Generator (PSG; G. L. Villanueva et al. 2018) and its radiative transfer tool for spectra synthesis, which we extend to use in a distributed computing setting. Data generation involved a large number of PSG runs and necessitated access to substantial computational resources. We used Google Cloud services for code execution and data storage in a distributed setting. The aspects of containerization and deployment of multiple PSG servers across virtual machines and the overall framework of distributed planetary system sampling, spectra simulation, and saving the resulting data into a unified data set are important aspects of the work that we are reporting.

The generated data set enables various ML approaches such as clustering, classification, regression and generative modeling of exoplanetary spectra. In this work we demonstrate a use case in supervised learning and report the results of a regression task to predict exoplanetary parameters from observed spectra using convolutional neural networks (CNNs). To enhance the reliability of our predictions and provide a measure of uncertainty, we implemented the Monte Carlo dropout approximation (Y. Gal & Z. Ghahramani 2015). Other potential uses for this data set include investigating the detectability of molecules in terrestrial atmospheres as well as investigating how well pressure–temperature profiles can be constrained for reflected light spectra of terrestrial exoplanets for some assumed telescope properties and noise models.

This paper focuses on the release and significance of the generated data set, emphasizing its relevance for advancing atmospheric retrieval methodologies on rocky exoplanets, while providing a concise overview of the ML framework as a contextual use case.

3. Methods

Here we describe the model we developed for the generation of planetary systems and corresponding spectra. In order to simulate a large number of exoplanetary spectra using PSG, we need to define a probability model of planetary systems from which we can draw samples and then simulate the corresponding reflected light spectra. This involves defining a 28-dimensional parameter space including the stellar and planetary parameters as well as the molecular abundances in the planet’s atmosphere, which we present in Table 1. We introduce a probability distribution over this parameter space and draw

Table 1
Summary Table of All Parameters of the Simulated Exoplanetary Systems

Index	Parameter	Description [Units]	Distribution or Derivation
0	Planet Index	...	...
1	Stellar Type	Stellar type	Uniform distribution over {F, G, K}
2	Stellar Temperature	Stellar temperature [kelvin]	Uniform distribution based on T. S. Boyajian et al. (2012) and T. S. Boyajian et al. (2013). See Table 2
3	Stellar Radius	Stellar radius [$R_{\odot}$]	Uniform distribution based on T. S. Boyajian et al. (2012) and T. S. Boyajian et al. (2013). See Table 2
4	Distance	Distance from Earth to system [parsecs]	Uniform distribution [1.3, 15] pc for coronagraphic observations (F, G, and K types)
5	Semimajor Axis	Semimajor axis of the planet [astronomical units]	From an optimistically habitable Venus to Mars scaled according to the host star's radius and temperature (R. K. Kopparapu et al. 2013)
6	Planet Radius	Planet radius [kilometers]	Uniform distribution [3189, 10,205] km, equivalent to [0.5, 1.6] Earth's radius
7	Planet Density	Planet density [g cm^{-3}]	Uniform distribution based on C. Sotin et al. (2007)
8	Planet Surface Pressure	Planet surface pressure [bar]	Truncated normal distribution with mean 1, s.d. 2.5, and bounds [0.1, 90]
9	Kappa	Thermal profile parameter of M. R. Line et al. (2013); Planck mean infrared opacity	Log-uniform distribution over $[-3.5, -2.0]$
10	Gamma1	Thermal profile parameter of M. R. Line et al. (2013); Planck mean visible to infrared opacity ratio for stream 1	Log-uniform distribution over $[-1.5, -1.1]$
11	Gamma2	Thermal profile parameter of M. R. Line et al. (2013); Planck mean visible to infrared opacity ratio for stream 2	Log-uniform distribution over $[-1.5, 0]$
12	Alpha	Thermal profile parameter of M. R. Line et al. (2013); relative contribution of the two streams	Uniform distribution over $[0, 1]$
13	Beta	Thermal profile parameter of M. R. Line et al. (2013); general term to represent albedo, emissivity, and energy recirculation	Truncated normal distribution with mean 0.95, s.d. 0.1, and bounds $[0.7, 1.1]$
14	Planet Surface Temperature	Planet surface temperature [kelvin]	Determined from $T(p)$ profile
15	H ₂ O Abundance	H ₂ O abundance	Uniform distribution; upper limit of 0.1
16	CO ₂ Abundance	CO ₂ abundance	Uniform distribution; upper limit of 1
17	O ₂ Abundance	O ₂ abundance	Uniform distribution; upper limit of 1
18	N ₂ Abundance	N ₂ abundance	Uniform distribution; upper limit of 1
19	CH ₄ Abundance	CH ₄ abundance	Uniform distribution; upper limit of 0.1
20	N ₂ O Abundance	N ₂ O abundance	Uniform distribution; upper limit of 0.2
21	CO Abundance	CO abundance	Uniform distribution; upper limit of 0.2
22	O ₃ Abundance	O ₃ abundance	Uniform distribution; upper limit of $0.01 \times \text{O}_2$
23	SO ₂ Abundance	SO ₂ abundance	Uniform distribution; upper limit of 0.02
24	NH ₃ Abundance	NH ₃ abundance	Uniform distribution; upper limit of 0.01
25	C ₂ H ₆ Abundance	C ₂ H ₆ abundance	Uniform distribution; upper limit of $0.01 \times \text{CH}_4$
26	NO ₂ Abundance	NO ₂ abundance	Uniform distribution; upper limit of 2×10^{-5}
27	Avg Mol Weight	Planetary atmosphere's average molecular weight [g mol^{-1}]	Calculated from the molecular abundances
28	Mean Surface Albedo	Planet's mean surface albedo [unitless]	Uniform distribution $[0.1, 0.8]$

samples from this distribution. We then simulate the corresponding spectra using PSG. The details of the parameter space, the probability distributions, and the data generation setup are described in the following sections.

3.1. Planetary System Parameters

We consider F, G, and K main sequence stars. Kurucz stellar models were used for G and K types, while F types are simulated as a blackbody, as PSG lacked an F-type model at the time of data generation. M-type stars could not be observed with a coronagraph due to how close the planets orbit the star, so were excluded during the data generation phase. INARA is also configured to simulate transiting exoplanets around M-type stars, thus this star type might be included into future

simulations. Stellar radii and temperatures are randomly selected from uniform ranges based on T. S. Boyajian et al. (2012, 2013) and the specific bounds for each stellar type (F, G, and K) are provided in Table 2. The semimajor axis of the planet is randomly selected from that of an optimistically habitable Venus to an optimistically habitable Mars scaled according to the host star's radius and temperature, as defined in R. K. Kopparapu et al. (2013). The distance of the system is randomly selected from 1.3 to 15 pc for coronagraphic observations. These ranges were chosen for observational reasons. For the minimum, the closest exoplanet is Proxima Centauri b at 1.3 pc (coronagraphic), and detectors will saturate quickly at 5 pc if staring at the target (transit). For the maximum, noise necessitates many hours of observations, reducing the likelihood that rocky worlds will be studied at

Table 2
Stellar Temperature and Radius Ranges Defining the Bounds of Continuous Uniform Distributions Modeling Each Stellar Type

Stellar Type	Temperature Range	Radius Range
F	5900–7200 K	$T > 6400$ K: $1.4\text{--}2.0 R_{\odot}$, $T \leq 6400$ K: $1.1\text{--}2.0 R_{\odot}$
G	5300–5900 K	$T > 5500$ K: $0.9\text{--}1.3 R_{\odot}$, $T \leq 5500$ K: $0.8\text{--}1.1 R_{\odot}$
K	3800–5300 K	$T > 5000$ K: $0.7\text{--}0.95 R_{\odot}$, $T \leq 5000$ K: $0.6\text{--}0.85 R_{\odot}$

Table 3
Upper Bounds of Continuous Uniform Distributions Used to Model Molecular Abundances in Planetary Atmospheres

Molecule	H ₂ O	CO ₂	O ₂	N ₂	CH ₄	N ₂ O	CO	O ₃	SO ₂	NH ₃	C ₂ H ₆	NO ₂
Upper bound	0.1	1.0	1.0	1.0	0.1	0.02	0.02	$0.01 \times \text{O}_2$	0.02	0.01	$0.01 \times \text{CH}_4$	2×10^{-5}

Note. The lower bound is zero for all molecules.

these distances given there are likely better targets at closer distances.

Observations are simulated assuming a 15 m space telescope with a spectral resolution of 1900 covering 0.2 to 640 μm . At $\leq 2 \mu\text{m}$, we assume there is a coronagraph that suppresses the stellar signal by a factor of 10^{10} . In addition to the combined stellar + planetary noise-free signal, we separately provide the stellar and planetary spectra as well as an example noise model with low read noise ($1 \text{ e}^- \text{ pixel}^{-1}$) and dark current ($0.001 \text{ e}^- \text{ s}^{-1}$). The coronagraph’s inner working angle is $2\lambda/D$. These are based on the Large Ultraviolet Optical Infrared Surveyor design concept, but with a much higher resolution and wider wavelength range to allow the use of both high- and low-resolution data for experimentation beyond what is considered here. At $> 2 \mu\text{m}$, we do not include a coronagraph nor carry out noise simulations. Consequently, the total, stellar, and noise spectra feature a discontinuity at $2 \mu\text{m}$. Simulated observations have similar noise characteristics; the nominal observations are 8 hr on Earth at 5 pc when it is at its greatest separation from the Sun. The observing times are modified by the squared distance ratio, such that a planet at twice the distance of the nominal case will be observed for 4 times longer. The total signal provided in the data set refers to the combined star plus planet spectra. A given noise spectrum is the signal (due to read noise, dark current, white noise, etc.) that would be added to that total noise-free signal if that system were observed by an instrument with the assumed noise characteristics.

3.2. Planetary Parameters

Planetary radii are randomly selected from a uniform range spanning 0.5 to $1.6 R_{\oplus}$ due to planets with radii $> 1.6 R_{\oplus}$ containing a significant gas mass fraction (L. A. Rogers 2015). Planetary masses are determined using the mass–radius relation of C. Sotin et al. (2007) with an adjustment to account for model inaccuracies drawn from a uniform distribution between $\pm 2\%$. Using a rough approximation of the “cosmic shoreline” model of K. J. Zahnle & D. C. Catling (2017), we throw out any planet that is likely unable to hold onto an atmosphere. We draw the surface pressure from a truncated normal distribution bound by 0.1 and 90 bars with a mean of 1 bar and a standard deviation of 2.5 bars. The pressure–temperature profile is determined from the parameterized formulation of M. R. Line et al. (2013). Four of these parameters, which govern the shape of the profile, are selected from a random log-uniform distribution, while the fifth, which shifts the profile in

temperature, is selected from a truncated normal distribution. The ranges for these parameters were chosen such that a planet in an Earthlike orbit can vary in temperature by a few hundred kelvin. As a result, both Venus-like planets and frozen Mars-like planets are within our parameter space. Our hottest case results in temperatures slightly below that of Venus. We allow the tropopause to vary between 0.023 and 0.23 bars due to the finding of T. D. Robinson & D. C. Catling (2014) that the planets in the solar system all have a tropopause around 0.1 bars.

We consider 12 molecules: H₂O, CO₂, O₂, N₂, CH₄, N₂O, CO, O₃, SO₂, NH₃, C₂H₆, and NO₂. Table 3 summarizes the upper bound on the random uniform distribution used to draw a value for each molecule; all molecules have a lower bound of 0 to allow for the possibility of negligible presence for certain species. Following the selection of these values, they are normalized such that they sum to 1; this yields the molar mixing ratio of each gas. As a result, it is possible for some of the trace species to exceed their “upper limit” if a high value is generated for the trace gas and low values are generated for the main constituents (CO₂, O₂, N₂). Our upper limits were selected to encapsulate as many atmospheres as possible while also restricting the parameter space to cases that are more likely for terrestrial planets. CO₂ and N₂ have been observed in excess of 90% in atmospheres in the solar system, hence their upper limits. O₂ is unlikely to exist at such high abundances but does exist at a substantial amount in Earth’s atmosphere; thus, we assume a large amount to allow for a wide range of possibilities. On Earth, H₂O content in the air can be a few percent in tropical regions; we allow for up to 10% to allow for conditions slightly more extreme than those found on Earth. O₃, a photochemical product of O₂, has been shown to be at most $\sim 1\%$ the amount of O₂ in prebiotic Earthlike atmospheres (S. D. Domagal-Goldman et al. 2014). We impose a similar limit for C₂H₆ as it is a photochemical product of CH₄. N₂O can act as a strong biosignature in certain situations, such as that of Earth, but must be considered in the context of the host star and the absence of gases indicative of abiotic N₂O production (E. W. Schwieterman et al. 2018). The upper limits on other trace gases are arbitrarily determined to cover many cases for thin atmospheres.

All gases begin with a uniform vertical abundance profile. It is important to note that while some gases (e.g., O₃) considered in our model may not be well mixed throughout the atmosphere under the conditions examined, we adopt the assumption of uniform abundances for practical reasons. Introducing a more complex parameterization for these gases would significantly

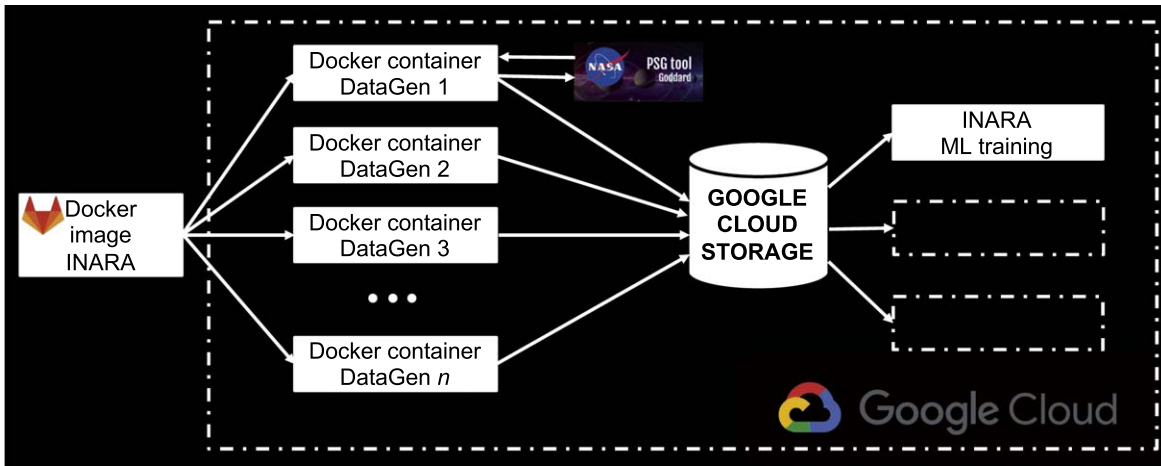


Figure 1. Overview of the distributed data generation and training setup. Once instantiated as a Docker container in the cloud, INARA communicates with a designated PSG VM to generate the data and store it in the cloud. INARA Docker containers are then able to read the data back in for ML training and inference purposes.

increase the total number of free parameters or require strict assumptions that may not be universally valid. Despite its nonphysical nature for certain gases, this approach aligns with many retrieval models in the literature. Importantly, it enables the evaluation of the accuracy of those retrieval models when applied to the generated data set. For H_2O and NH_3 , we modify their abundance profiles by calculating the saturation vapor pressure (SVP) at each layer in the atmospheric model. We assume that all excess vapor pressure condenses out into clouds. The SVPs are calculated using the Antoine equation, leveraging available NIST data. Specifically, H_2O covers a range of 255.9–573 K, while NH_3 covers a range of 164–371.5 K. On an Earthlike planet, this method results in the abundance of H_2O decreasing with increasing altitude, reaching a cold trap beyond which clouds no longer form. While we calculate cloud mixing ratios, the simulations exclude the consideration of clouds due to computational constraints. Even modest attempts to model clouds increased computational time by a factor of ~ 50 at the time of conducting the simulations. This limitation serves as an upper bound on any retrieval constraints when using the present data set, as clouds mute spectral features and thus generally lead to weaker constraints on atmospheric properties. As such, our data set can be used to determine regions of the parameter space where certain parameters cannot be constrained even in the cloud-free case, allowing those regions to be excluded from more computationally expensive studies that consider clouds. It is worth noting that users have the flexibility to selectively reprocess certain cases with scattering included to capture the full impact of clouds. Additionally, future enhancements in the efficiency of scattering calculations may facilitate reprocessing the entire data set with the inclusion of scattering effects from clouds within a reasonable runtime. This optional reprocessing step underscores the adaptability of our framework.

3.3. Data Generation, Computational, and Software Environment

The computational demands of running a large number of PSG simulations necessitated a distributed computing approach in order to generate the data set in a reasonable timeframe. In order to deploy PSG at scale, we developed the INARA framework. INARA is a unified codebase for instantiating

virtual machines (VMs) on the Google Cloud platform, running Docker containers with PSG simulation servers, sampling planetary systems and requesting the corresponding spectra from PSG servers, saving the generated data in cloud storage, and reading the data back in for ML training and inference purposes (Figure 1). We use INARA to instantiate 2000 VMs running PSG. For each PSG node, we instantiate 16 worker VMs, which draw samples from the parameter distributions that we defined (Table 1), submit requests to the PSG node via the `pypsg` Python package that we implemented, and store the spectra in cloud storage. Once data generation is complete, INARA can use the generated data to train ML models via the PyTorch library.

INARA can be run for data generation, ML model training, and model inference. The code runs locally on a system depending on installed software requirements (the GitLab repository includes a conda environment file with all requirements) or in the form of a Docker container. The latter option requires no installation of software dependencies or ML frameworks as the container can be deployed to run on any Docker-supported operating system. While INARA is developed to run PSG simulations on the Google Cloud platform, the general methodology can in theory be applied to any code on any cloud platform.

The generated data set (INARA DS 1) contains 3,112,620 synthetic exoplanetary systems. The simulations for each system are stored within a CSV file, which contains four vectors: the total (star + planet) spectrum, the simulated noise, the stellar spectrum, and the planet spectrum. Each vector contains 15,346 elements, covering wavelengths from 0.2 to $640\ \mu\text{m}$, and all values are in $\text{erg s}^{-1} \text{cm}^{-2}$. Data are available as individual files and in compressed archives (see Section 4 for details). An associated summary file contains the input parameters for each system.

4. Data Availability

Since the INARA data set contains data on exoplanets and host stars, the Exoplanet Archive is an ideal choice for a repository for the public release. The Exoplanet Archive provides tools to search, visualize, and analyze data for planetary and stellar physical and orbital properties (R. Akeson et al. 2013). The Exoplanet Archive hosts data associated with exoplanets, such as measured and derived parameters, spectra,

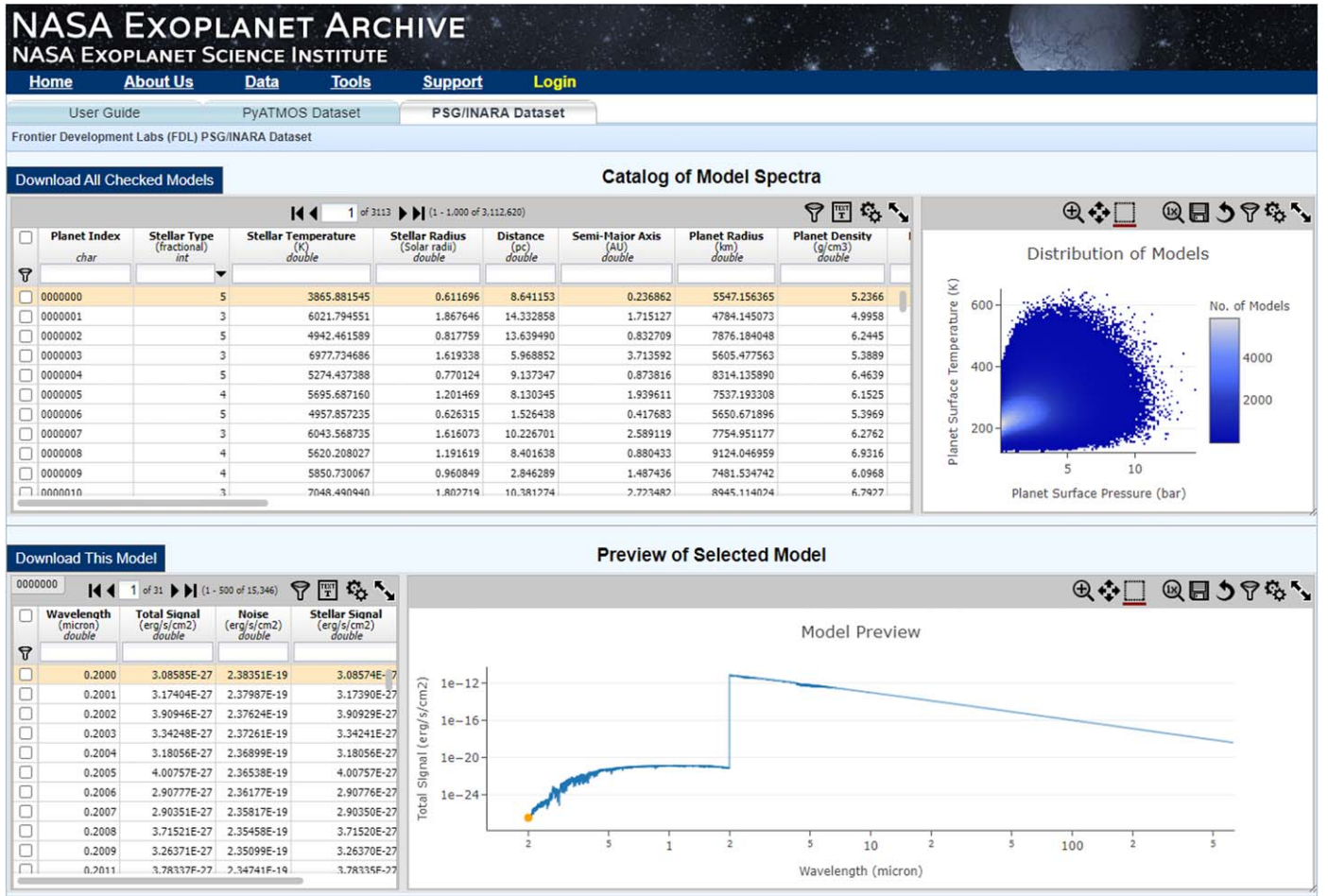


Figure 2. INARA data set Exoplanet Archive web interface: For each spectrum, the user can access the parameters used to generate the data (top panel), a graphical interface with a distribution of spectra across the space of the parameters (top-right plot), and plots for the selected spectrum (bottom-right plot). The table and plot interfaces provide many more functionalities via the icons in the gray bar at the top of each. For more information, refer to the documentation page at the Exoplanet Archive.

and time series. These data sets come from the core archive and from data sets provided by groups conducting searches for exoplanets. Among these sets are high-resolution spectra from the N2K Consortium (D. A. Fischer et al. 2007), the California Planet search team (J. Wright et al. 2012), the M2K program (D. A. Fischer et al. 2011), and the NASA Frontier Development Lab program, which includes the PyAtmos (A. Chopra et al. 2020) data set and the INARA data set (NASA Exoplanet Archive 2023).

The INARA data set is available via an interface (Figure 2) for interactively browsing of the data set at <https://exoplanetarchive.ipac.caltech.edu/cgi-bin/FDL/nph-fdl?psg>.¹⁰ A user guide is available at the corresponding link in the upper-left corner. The upper panel displays a summary table named *Catalog of Model Spectra* of all the available synthetic spectra, with one row per spectrum, and with columns representing the input parameters for each model spectrum, as described above. By default, the plot to the right of the *Catalog of Model Spectra* table shows a two-dimensional histogram of the distribution of the simulated spectra across input parameter space, initially against planet surface pressure and planet surface temperature. The bottom panel shows a plot of the individual synthetic spectrum

indicated by the currently selected row in the summary table above. The table in the bottom panel, named *Preview of Selected Model*, shows the actual data content available for the spectrum, including the separated planet, star, and noise components alone, as well as the total star + planet spectrum (see the “Data Descriptions” section on the Exoplanet Archive website for more information).¹¹ Clicking on any row in the *Catalog of Model Spectra* summary table (top panel) highlights that row and updates the model preview panel (bottom panel) accordingly. The selected row in the bottom table is indicated by a highlighted point on the curve in the spectrum preview plot to the right. Clicking any row in the lower table selects the associated point in the plot, and vice versa. Both of the tables can be sorted by clicking on the column headings, and filtered by entering filter criteria into the text boxes underneath the column headings (e.g., the user might enter “>300” in the “Planet Surface Temperature” column of the summary table). The respective plots will be updated accordingly.

4.1. Downloading Data

Downloading a single synthetic spectrum. To download the data for a single model, the user can highlight the row

¹⁰ Since the data set is large, please allow a few minutes for the server to reload if it has not been used recently.

¹¹ <https://exoplanetarchive.ipac.caltech.edu/cgi-bin/FDL/nph-fdl?psg>

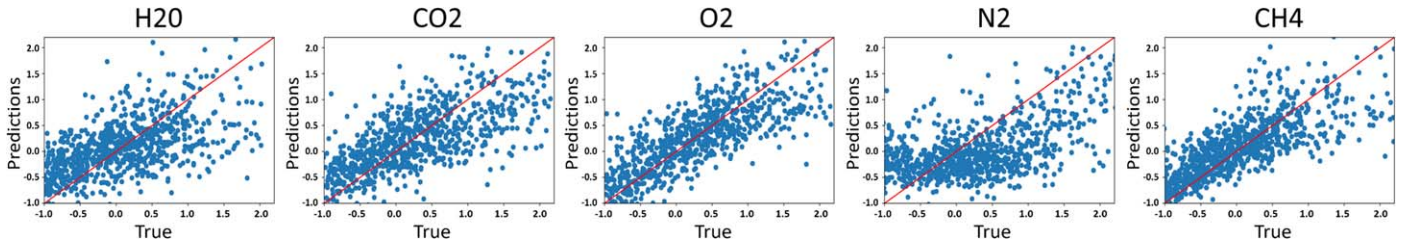


Figure 3. INARA prediction performance for the logarithm of the normalized abundance of H_2O , CO_2 , O_2 , N_2 and CH_4 across 1000 test planets. We normalize values by subtracting the mean and dividing by the standard deviation across all planets. Each dot is the average of 600 runs with dropout for a single planet. Predicted vs. true values are plotted, with the diagonal line indicating perfect correspondence.

corresponding to the model in the summary table in the top panel, and then click “Download This Model” in the bottom panel. This will download a single CSV file that contains the same data as displayed in the lower table to the user’s local drive.

Downloading a selection of synthetic spectra. To download data for a selection of spectra, the user can indicate which of the spectra will be downloaded by checking their respective checkboxes. The filters on the columns or the summary allow to make a multiple selection, while the “Download All Checked Models” will generate a script that can be run by the user to retrieve all the required spectra, up to 10,000. For larger downloads, see the next section as well as the batch scripts available at the NASA Exoplanet Archive.

Downloading the entire data set. To download the entire set of over three million synthetic spectra, a collection of scripts are available through the NASA Exoplanet Archive page for the INARA data set project. These data sets can include the data of all stellar spectra, planetary spectra, noise spectra, stellar and planetary spectra combined, or all the parameters used for the spectra generation (complete summary table). These downloads vary in size from 862 MB (2.3 GB uncompressed) to 269 GB compressed (782 GB uncompressed). For a description of the files delivered with each model, and download sizes see “Data Descriptions” in the project web page.

5. Example Use Case: Machine Learning Atmospheric Retrieval

Using a subset of the data created by running the PSG simulator with the parameters sampled as described in Section 3, we train a deep neural network (NN) in a supervised setting to predict exoplanet atmospheric characteristics given an observed spectrum. In order to train and evaluate the NN model, we use a training set of 110,000 cases, a validation set of 10,000 cases, and a test set of 1000 cases. Spectra were represented as vectors of length 4379 that describe radiation intensity as a scalar function of wavelengths less than $2\ \mu\text{m}$.

We performed a model grid search to determine an NN architecture well suited to solve this problem. We consider linear regression, feed-forward, and CNN models; learning rates in $[0.0001, 0.01]$; ADAM, stochastic gradient descent, ADAdelta, and RMSProp optimizers; and tanh, rectified linear unit, exponential linear unit, and linear activation functions. Using a mean-squared-error loss, we train each model for 64 epochs using early stopping to prevent overfitting. We choose the optimal model architecture based on the minimum validation loss. Based on the grid search results, we find that 1D CNNs yield the best results. For this problem, we found an optimal model configuration of *Conv1d(64)-tanh-MaxPool-Conv1d(64)-relu-MaxPool-Conv1d(128)-relu-MaxPool-*

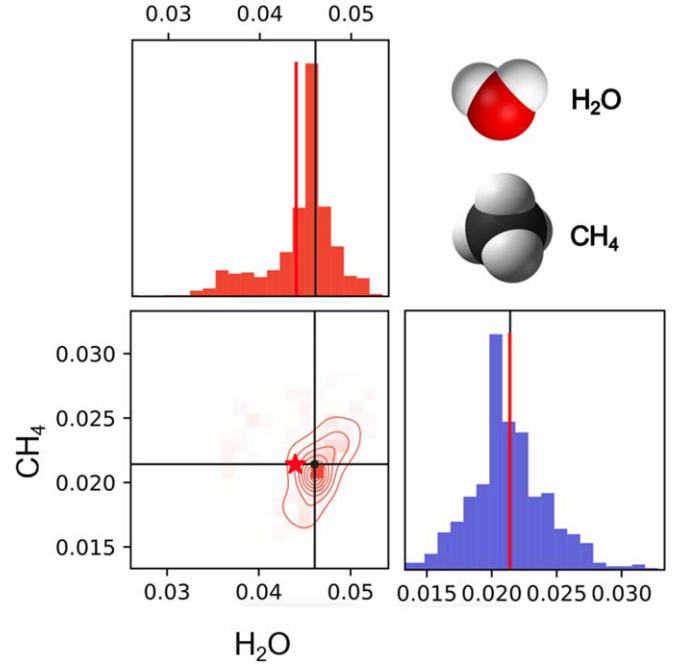


Figure 4. Detailed results of INARA in predicting one random planet’s values for H_2O and CH_4 . The distribution is obtained by 600 predictions with dropout where nodes were randomly removed from the network. The black lines represent INARA’s median prediction, and the red lines and star represent, respectively, the real value in the dimensions of H_2O and CH_4 , and in the two-dimensional space with one molecule per axis.

Conv1d(256)-relu-FC(256)-relu-FC(12), which has approximately 18 million trainable parameters.

When training and evaluating the optimal model, we utilize Monte Carlo dropout approximation, which makes a variational approximation to the posterior distribution over the network weights (Y. Gal & Z. Ghahramani 2016). Each dropout mask removes a certain proportion, p , of NN weight connections by setting them to zero during a forward pass. Therefore, multiple forward passes with different dropout masks for the same input gives a set of predictive samples that build a predictive distribution. Through implementing dropout both at training and test time, we are effectively sampling from the posterior over weights of the network. This distribution over the weights enables us to approximate a predictive distribution over each model parameter, which we represent as the output samples of the network for a given input.

Figure 3 shows the mean prediction compared to the true value for each planet in the test set, with the red line indicating a perfect prediction as the identity function.

Figure 4 shows a predictive distribution over H_2O and CH_4 abundances resulting from dropout for a specific test planet.

This example is intended solely as a proof of concept to demonstrate the feasibility of using the data set for ML-based atmospheric retrieval. We emphasize that the trained NN's predicted distributions generally encapsulate the true values, but the scope of this demonstration is limited to a small subset of the data set and noiseless spectra for simplicity. While training on the full data set and incorporating realistic noise models would likely improve the results and broaden their applicability, such an extension is beyond the scope of this paper, which focuses primarily on presenting the data set itself. Detailed ML applications based on the complete data set are deferred to future work.

6. Summary and Future Work

In this paper we present a data set of 3,112,620 spectra of synthetic extrasolar systems containing terrestrial exoplanets in the optimistically habitable zone generated using the NASA PSG. We considered 28 free parameters describing the star-planet system, stellar properties, and planetary properties, encompassing terrestrial exoplanets with properties ranging from Venus-like to Mars-like. We also presented the INARA framework, which deployed the data generation via PSG at scale on the Google Cloud platform. Using INARA, we trained a deep neural network on a subset of the simulated data set to quickly perform an approximate atmospheric retrieval, demonstrating an example application of the data set. INARA is publicly available on Gitlab.¹²

The data set is publicly hosted at the Exoplanet Archive, offering a browsable interface to explore the data set, including visualizations and graphical representations of the data. Using the interface, users can filter, select, and download specific spectra. For larger computations, users can download the full data set or certain subsets of the data (e.g., only the planetary spectra) via scripts prepared by the Exoplanet Archive.

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¹² <https://gitlab.com/frontierdevelopmentlab/astrobiology/inara>

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